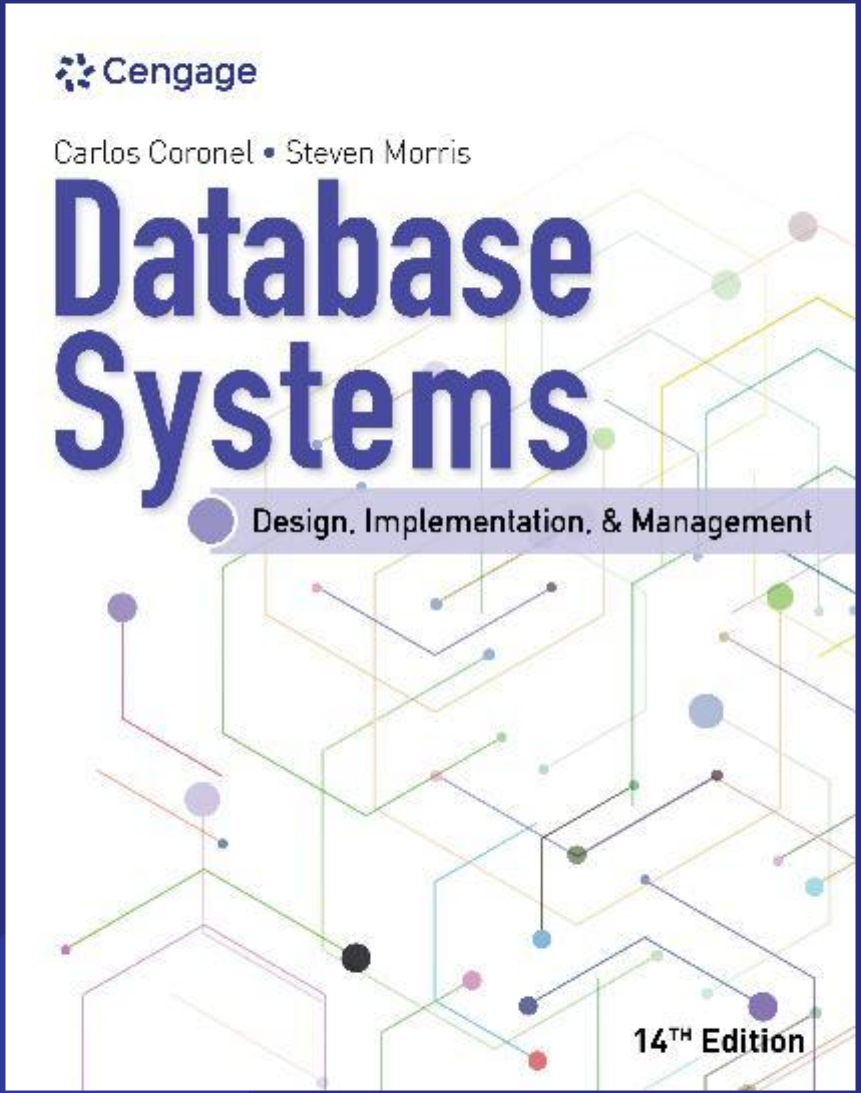


Database Systems: Design, Implementation, and Management, 14e

Data Models

Chapter Objectives

1. Discuss data modeling and why data models are important
2. Describe the basic data-modeling building blocks
3. Define what business rules are and how they influence database design

Data Modeling and Data Models

- **Data modeling** refers to the process of creating a specific data model for a determined problem domain
  - Data modeling is an iterative, progressive process
- A **data model** is a relatively simple representation of more complex real-world data structures
- Database designers use existing models and design tools to reduce errors in building databases.

The Importance of Data Models

- Data models are a communication tool
- Data models act as blueprints for understanding and using data.
- Different people (programmers, managers, end users) view data differently
- A clear data model makes sure everyone works from the same foundation
- A good database requires creating an appropriate data model first.

Data Model Basic Building Blocks

- An **entity** is a thing or object about which data is collected (e.g., a student, a course).
- An **attribute** is a characteristic or property of an entity (e.g., student name, course ID).
- A **relationship** is a link or association between entities (e.g., a student enrolls in a course).
- A **constraint** is a rule that defines valid data or relationships (e.g., a student can't enroll in two courses at the same time slot).

Entity

- An **entity** is a person, place, thing, concept, or event about which data will be collected and stored
- For example:
  - Customer
  - Agent

Attribute

- An **attribute** is a characteristic of an entity
- For example customer entity will be described as follows:
  - customer first name
  - customer last name
  - customer phone number
  - customer address
  - customer credit limit

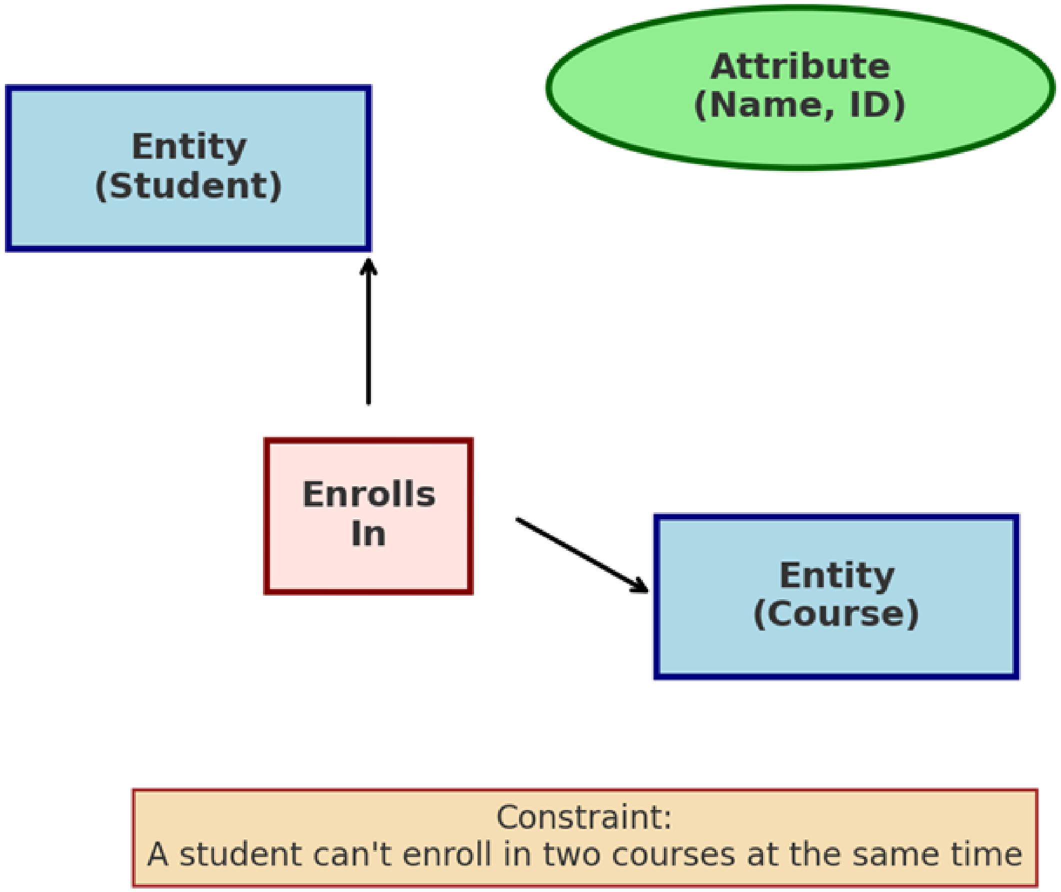
Relationship

- A **relationship** describes an association among entities.
- For example a relationship between a customer and an agent can be described as follows:
  - An agent can serve many customers
  - Each customer can be served by one agent

Constraint

- A **constraint** is a restriction placed on the data
  - Constraints help ensure data integrity
- Constraints are normally expressed in the form of **rules**:
  - An employee's salary must have values that are between 6,000 and 350,000
  - A student's GPA must be between 0.00 and 4.00
  - Each class must have one and only one teacher

Visual Diagram of the Basic Building Blocks of a Data Model



Types of Relationships

- The following are three different types of relationships:
  - **One-to-many (1:M or 1..*) relationship**
    - A painter creates many paintings.
    - "PAINTER paints PAINTINGS" (1:M)
  - **Many-to-many (M:N or *..*) relationship**
    - An Employee learns many job skills and each job skill may be learned by many employees.
    - "EMPLOYEE learns SKILL" (M:N)
  - **One-to-one (1:1 or 1..1) relationship**
    - Each store is managed by a single employee.
    - "EMPLYEE manages STORE" (1:1)

Knowledge Check Activity

- **What is a relationship, and what three types of relationships exist?**

Knowledge Check Activity: Answer

- **What is a relationship, and what three types of relationships exist?**
  - **Answer:** A relationship is an association among (two or more) entities. Three types of relationships exist: one-to-one (1:1), one-to-many (1:M), and many-to-many (M:N or M:M.)

Business Rules

- **Business rules** are clear statements of an organization’s policies or procedures.
- They apply to any organization that uses data for information.
- They should be simple and easy to use.
- Business rules are used to define entities, attributes, relationships, and constraints
- Example rules:
  - A customer can generate many invoices.
  - An invoice is generated by one customer.

Discovering Business Rules

- The main Sources of Business Rules are:
  - Company managers and policy makers
  - Department managers
  - Written documents (e.g., procedures, policies)
- Why Business Rules Matter in Database Design:
  - Standardize how the company views its data
  - Improve communication between users and designers
  - Help designers understand the data’s role and scope
  - Clarify business processes
  - Define relationships, rules, and constraints for accurate data models

Translating Business Rules into Data Model Components

- For example, the business rule "a customer may generate many invoices" contains two nouns and a verb that associates the nouns
- From this business rule, you could deduce the following:
  - Customer and invoice are objects of interest for the environment and should be represented by their respective entities
  - There is a generate relationship between customer and invoice
- The rule above is complemented by the business rule "an invoice is generated by only one customer"
  - The relationship is one-to-many (1:M)

Naming Conventions

- Entity names should be descriptive of the objects in the business environment and use technology that is familiar to the users
- An attribute name should also be descriptive of the data represented
- It is good practice to prefix the name of an attribute with the name or abbreviation of the entity in which it occurs
  - For example, in the CUSTOMER entity, customer’s credit limit may be called CUS_CREDIT_LIMIT
- A proper naming convention can help make your model self-documenting

The Relational Model

- The **relational model's** foundation is a mathematical concept known as a relation
  - A **relation** is a two-dimensional structure composed of intersecting rows and columns
  - Each row in a relation is called a **tuple** and each column represents an attribute
- The relational data model is implemented through a very sophisticated **relational database management system (RDBMS)**
  - The RDBMS performs the same basic functions provided by the hierarchical and network DBMS systems
- The RDBMS manages all of the details, while the users sees a collection of tables in which the data is stored

The Relational Model

Table name: AGENT (first six attributes)

Database name: Ch02_InsureCo

AGENT_CODE	AGENT_LNAME	AGENT_FNAME	AGENT_INITIAL	AGENT_AREACODE	AGENT_PHONE
501	Alby	Alex	B	713	228-1249
502	Hahn	Leah	F	615	882-1244
503	Okon	John	T	615	123-5589

Link through AGENT_CODE

Table name: CUSTOMER

CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_AREACODE	CUS_PHONE	CUS_INSURE_TYPE	CUS_INSURE_AMT	CUS_RENEW_DATE	AGENT_CODE
10010	Ramas	Alfred	A	615	844-2573	T1	100.00	05-Apr-2022	502
10011	Dunne	Leona	K	713	894-1238	T1	250.00	16-Jun-2022	501
10012	Smith	Kathy	W	615	894-2285	S2	150.00	29-Jan-2023	502
10013	Olowski	Paul	F	615	894-2180	S1	300.00	14-Oct-2022	502
10014	Orlando	Myron		615	222-1672	T1	100.00	28-Dec-2023	501
10015	O'Brian	Amy	B	713	442-3381	T2	850.00	22-Sep-2022	503
10016	Brown	James	G	615	297-1228	S1	120.00	25-Mar-2023	502
10017	Williams	George		615	290-2556	S1	250.00	17-Jul-2022	503
10018	Farriss	Anne	G	713	382-7185	T2	100.00	03-Dec-2022	501
10019	Smith	Olette	K	615	297-3809	S2	500.00	14-Mar-2023	503

Figure 2.1 Linking Relational Tables

The Relational Model

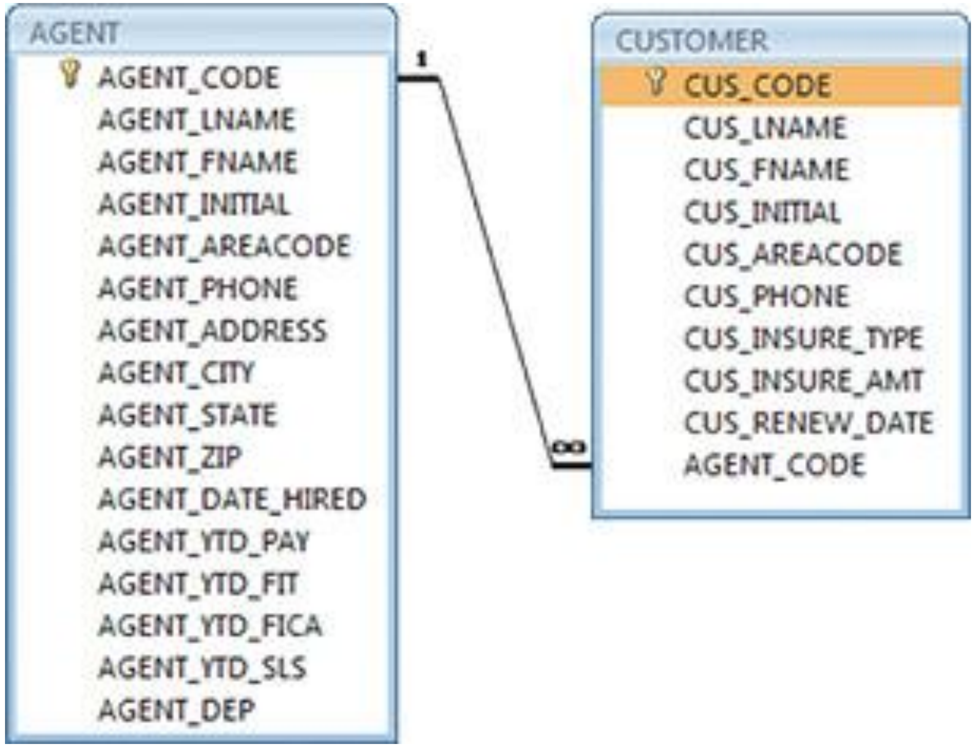


Figure 2.2 A Relational Diagram